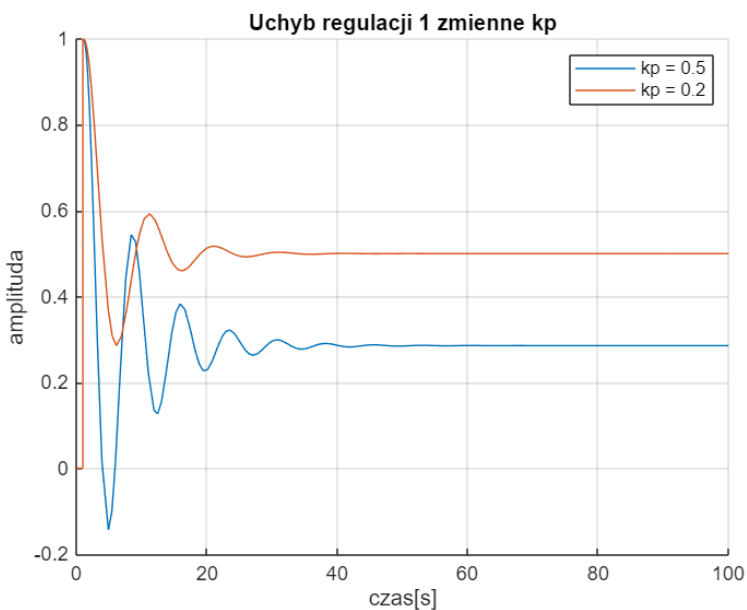


Sprawozdanie

Jakość regulacji

Jan Majerczyk

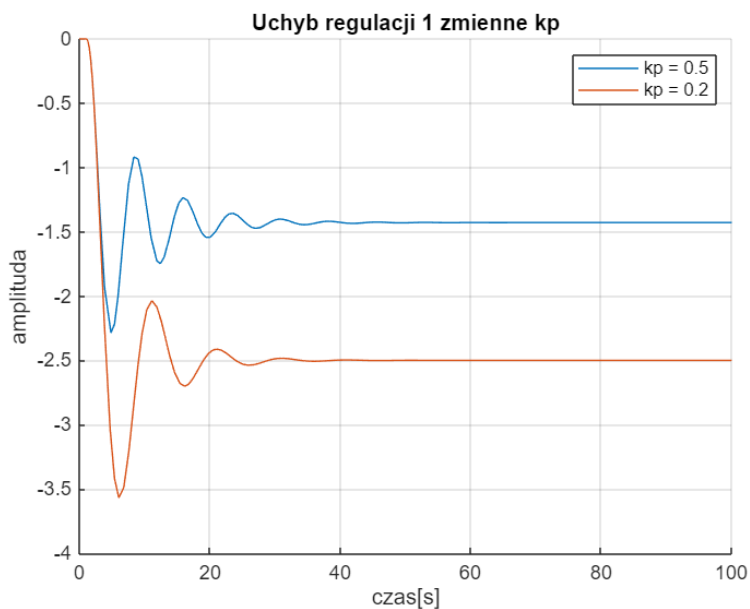
```
licznik1 = 5;  
mianownik1 = [1 5 2 1];  
licznik2 = 1;  
mianownik2 = [2 1 1 0];  
licznik = licznik1;  
mianownik = mianownik1;  
r = 1;  
z = 0;  
kp = 0.5;  
ki = 0;  
kd = 0;  
simOut = sim('regulacja.slx');  
err1 = simOut.get("err");  
kp = 0.2;  
ki = 0;  
kd = 0;  
simOut = sim('regulacja.slx');  
err2 = simOut.get("err");  
figure  
hold on  
plot(err1)  
plot(err2)  
hold off  
grid on  
title('Uchyb regulacji 1 zmienne kp')  
xlabel('czas[s]')  
ylabel('amplituda')  
legend('kp = 0.5', 'kp = 0.2')
```



```

r = 0;
z = 1;
kp = 0.5;
ki = 0;
kd = 0;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
kp = 0.2;
ki = 0;
kd = 0;
simOut = sim('regulacja.slx');
err2 = simOut.get("err");
figure
hold on
plot(err1)
plot(err2)
hold off
grid on
title('Uchyb regulacji 1 zmienne kp')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 0.5', 'kp = 0.2')

```



```

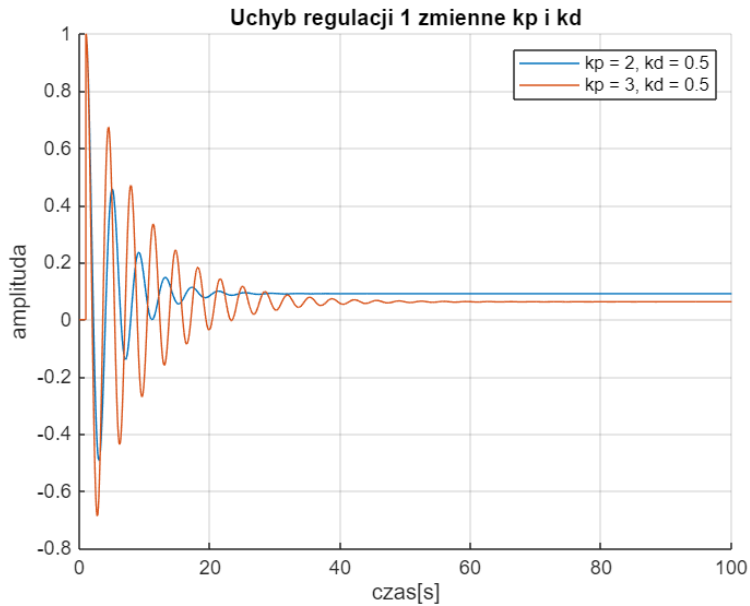
r = 1;
z = 0;
kp = 2;
ki = 0;
kd = 0.5;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
kp = 3;
ki = 0;
kd = 0.5;
simOut = sim('regulacja.slx');

```

```

err2 = simOut.get("err");
figure
hold on
plot(err1)
plot(err2)
hold off
grid on
title('Uchyb regulacji 1 zmienne kp i kd')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 2, kd = 0.5', 'kp = 3, kd = 0.5')

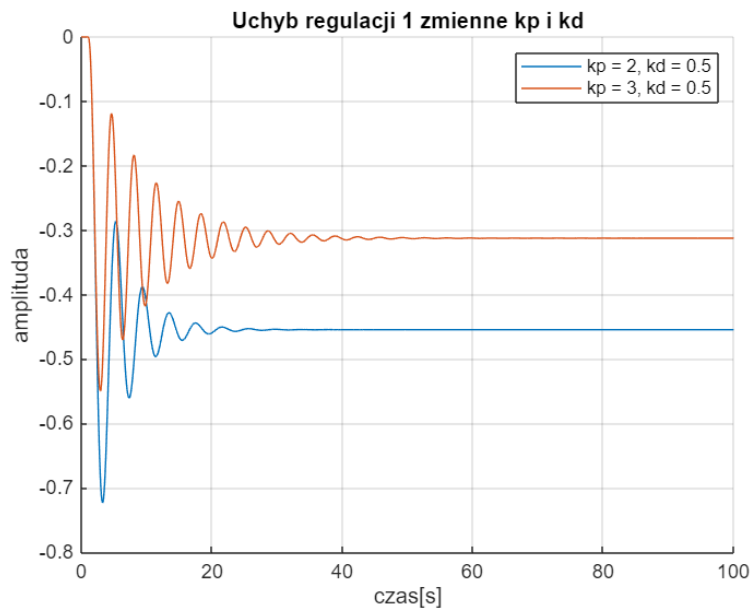
```



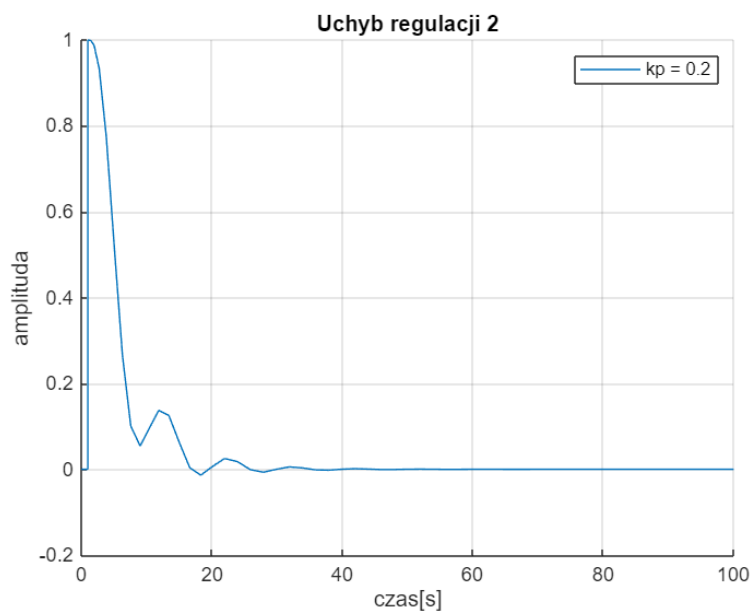
```

r = 0;
z = 1;
kp = 2;
ki = 0;
kd = 0.5;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
kp = 3;
ki = 0;
kd = 0.5;
simOut = sim('regulacja.slx');
err2 = simOut.get("err");
figure
hold on
plot(err1)
plot(err2)
hold off
grid on
title('Uchyb regulacji 1 zmienne kp i kd')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 2, kd = 0.5', 'kp = 3, kd = 0.5')

```



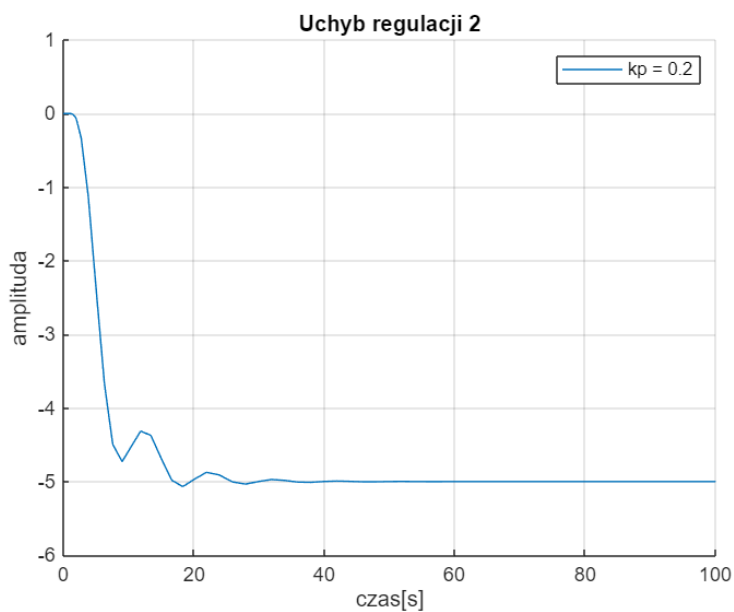
```
licznik = licznik2;  
mianownik = mianownik2;  
r = 1;  
z = 0;  
kp = 0.2;  
ki = 0;  
kd = 0;  
simOut = sim('regulacja.slx');  
err1 = simOut.get("err");  
figure  
hold on  
plot(err1)  
hold off  
grid on  
title('Uchyb regulacji 2')  
xlabel('czas[s]')  
ylabel('amplituda')  
legend('kp = 0.2')
```



```

r = 0;
z = 1;
kp = 0.2;
ki = 0;
kd = 0;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
figure
hold on
plot(err1)
hold off
grid on
title('Uchyb regulacji 2')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 0.2')

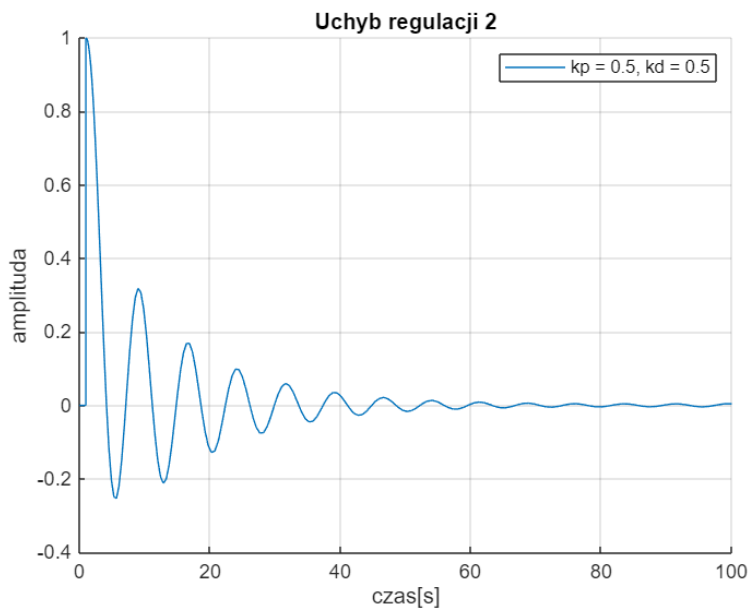
```



```

r = 1;
z = 0;
kp = 0.5;
ki = 0;
kd = 0.5;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
figure
hold on
plot(err1)
hold off
grid on
title('Uchyb regulacji 2')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 0.5, kd = 0.5')

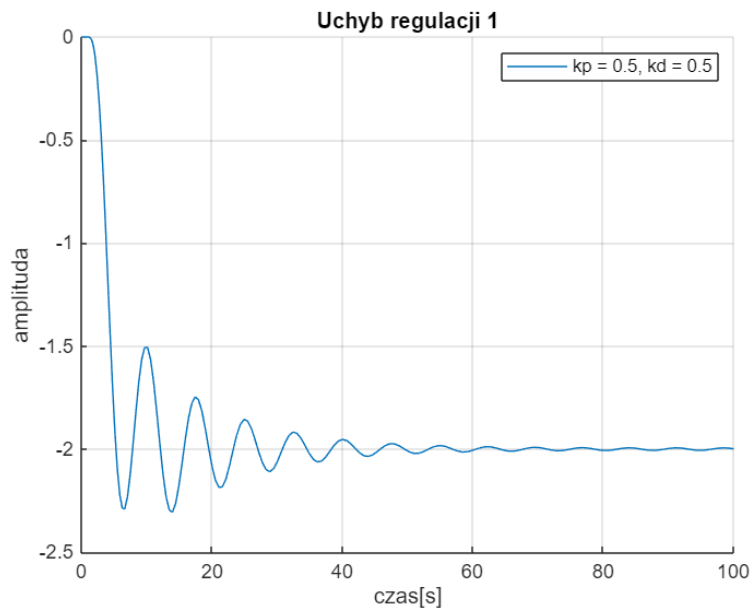
```



```

r = 0;
z = 1;
kp = 0.5;
ki = 0;
kd = 0.5;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
figure
hold on
plot(err1)
hold off
grid on
title('Uchyb regulacji 1')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 0.5, kd = 0.5')

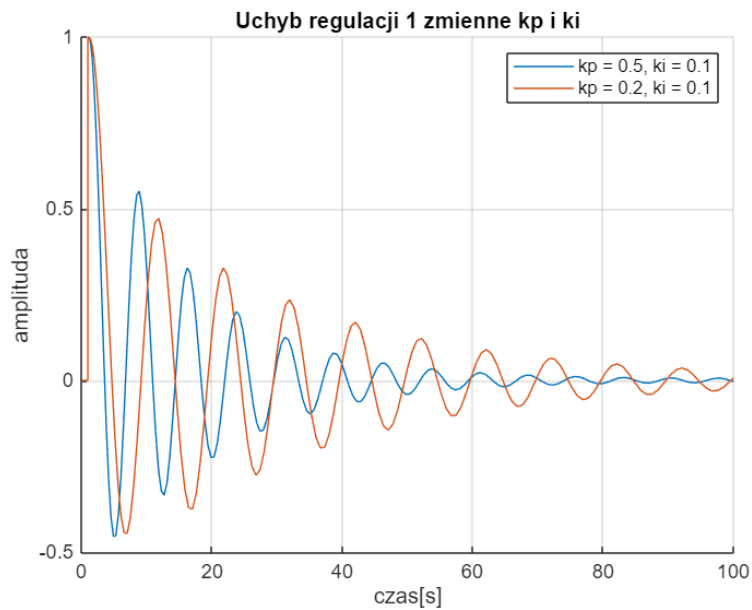
```



```

licznik = licznik1;
mianownik = mianownik1;
r = 1;
z = 0;
kp = 0.5;
ki = 0.1;
kd = 0;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
kp = 0.2;
ki = 0.1;
kd = 0;
simOut = sim('regulacja.slx');
err2 = simOut.get("err");
figure
hold on
plot(err1)
plot(err2)
hold off
grid on
title('Uchyb regulacji 1 zmienne kp i ki')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 0.5, ki = 0.1', 'kp = 0.2, ki = 0.1')

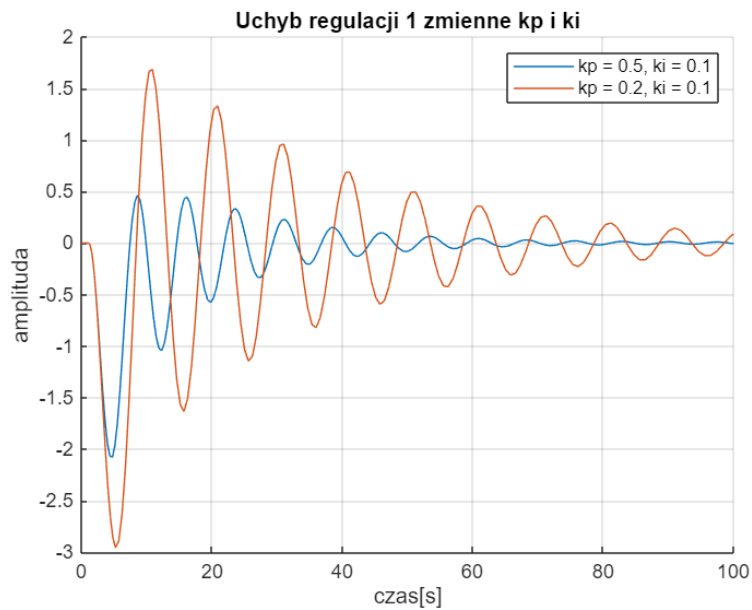
```



```

r = 0;
z = 1;
kp = 0.5;
ki = 0.1;
kd = 0;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
kp = 0.2;
ki = 0.1;
kd = 0;
simOut = sim('regulacja.slx');
err2 = simOut.get("err");
figure
hold on
plot(err1)
plot(err2)
hold off
grid on
title('Uchyb regulacji 1 zmienne kp i ki')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 0.5, ki = 0.1', 'kp = 0.2, ki = 0.1')

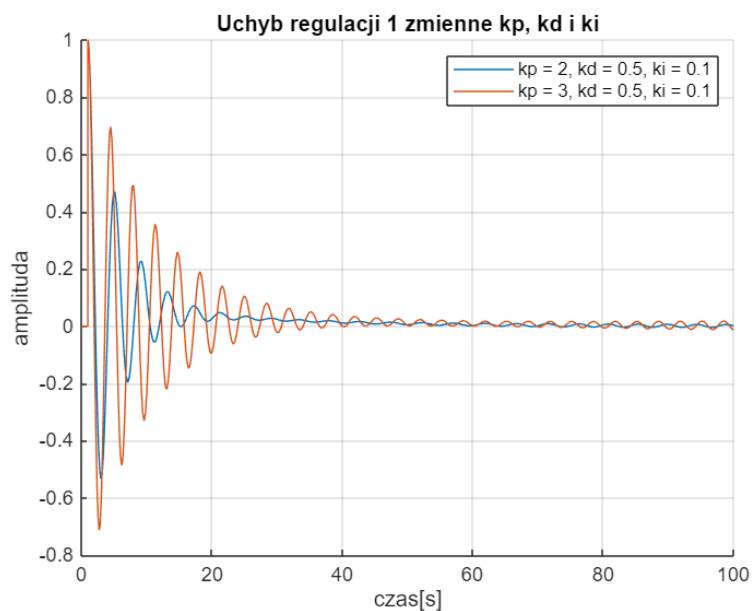
```



```

r = 1;
z = 0;
kp = 2;
ki = 0.1;
kd = 0.5;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
kp = 3;
ki = 0.1;
kd = 0.5;
simOut = sim('regulacja.slx');
err2 = simOut.get("err");
figure
hold on
plot(err1)
plot(err2)
hold off
grid on
title('Uchyb regulacji 1 zmienne kp, kd i ki')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 2, kd = 0.5, ki = 0.1', 'kp = 3, kd = 0.5, ki = 0.1')

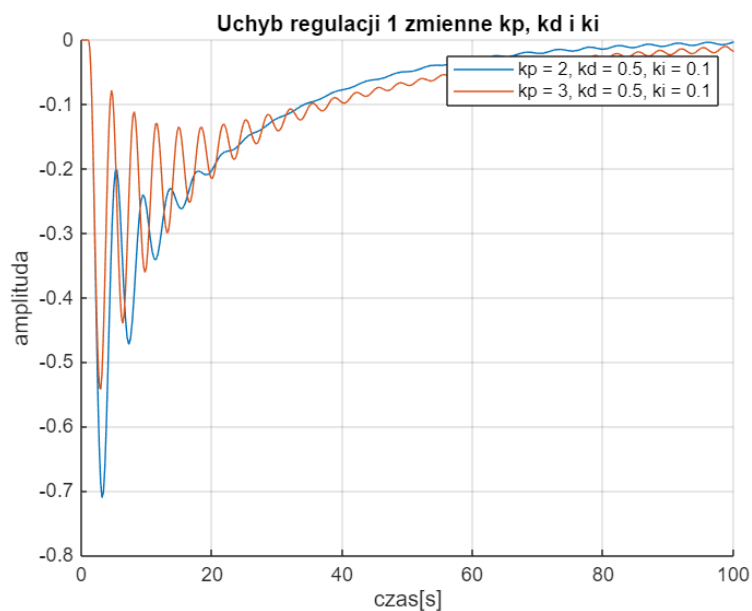
```



```

r = 0;
z = 1;
kp = 2;
ki = 0.1;
kd = 0.5;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
kp = 3;
ki = 0.1;
kd = 0.5;
simOut = sim('regulacja.slx');
err2 = simOut.get("err");
figure
hold on
plot(err1)
plot(err2)
hold off
grid on
title('Uchyb regulacji 1 zmienne kp, kd i ki')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 2, kd = 0.5, ki = 0.1', 'kp = 3, kd = 0.5, ki = 0.1')

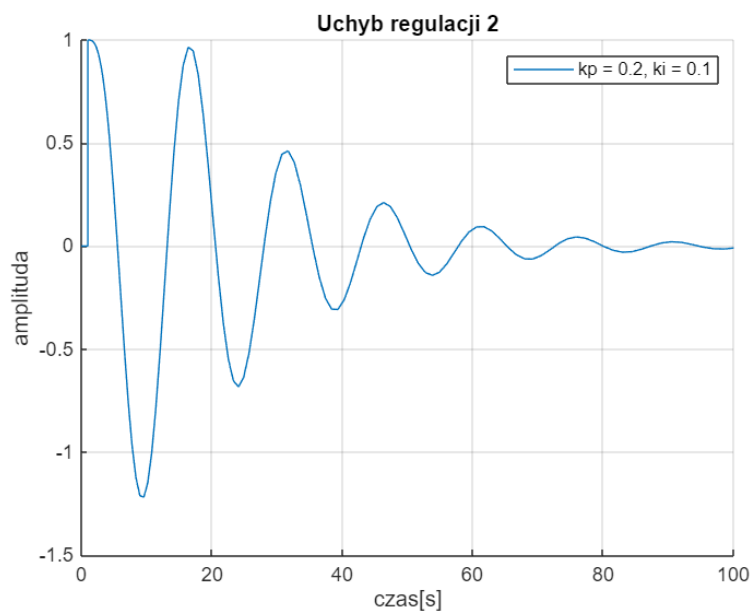
```



```

licznik = licznik2;
mianownik = mianownik2;
r = 1;
z = 0;
kp = 0.2;
ki = 0.1;
kd = 0;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
figure
hold on
plot(err1)
hold off
grid on
title('Uchyb regulacji 2')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 0.2, ki = 0.1')

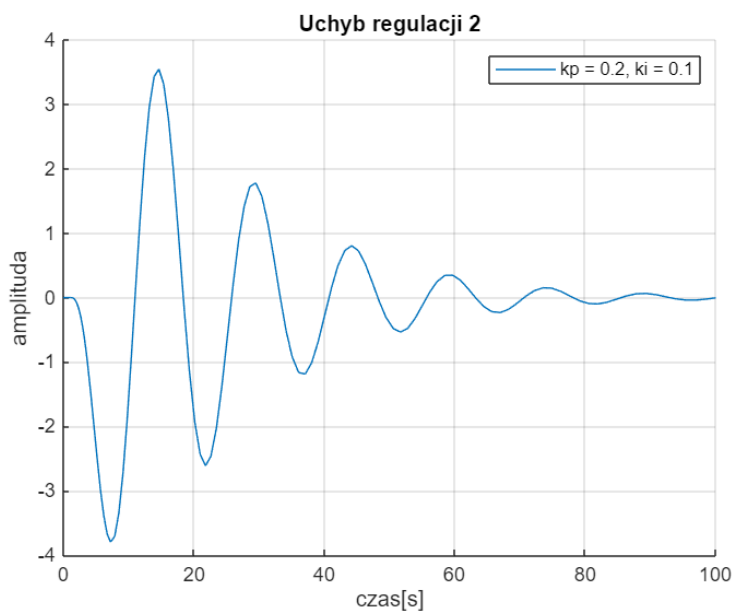
```



```

r = 0;
z = 1;
kp = 0.2;
ki = 0.1;
kd = 0;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
figure
hold on
plot(err1)
hold off
grid on
title('Uchyb regulacji 2')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 0.2, ki = 0.1')

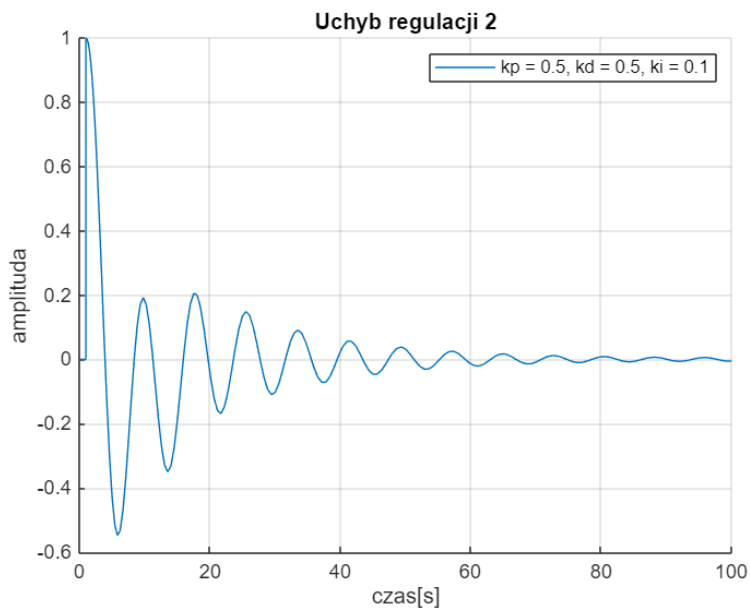
```



```

r = 1;
z = 0;
kp = 0.5;
ki = 0.1;
kd = 0.5;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
figure
hold on
plot(err1)
hold off
grid on
title('Uchyb regulacji 2')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 0.5, kd = 0.5, ki = 0.1')

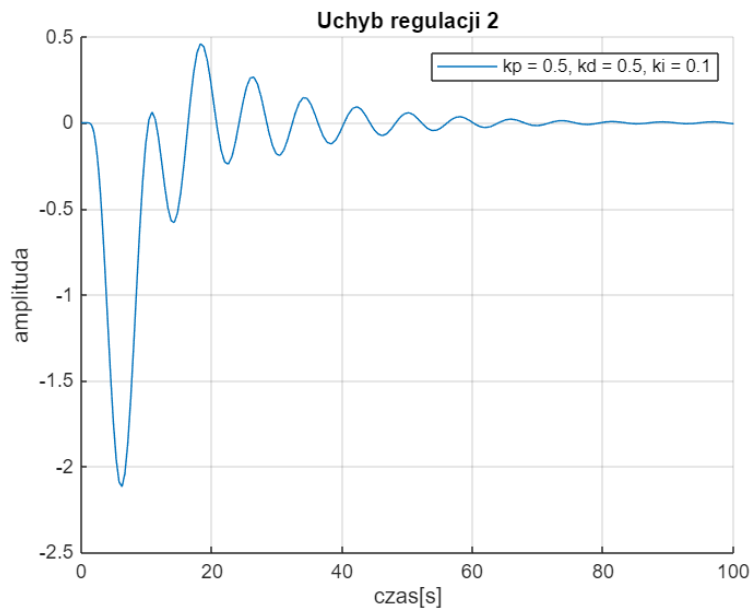
```



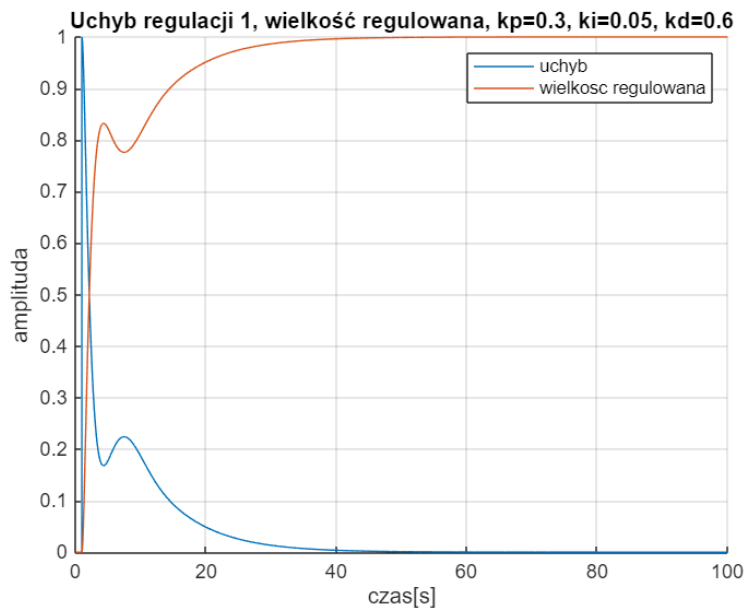
```

r = 0;
z = 1;
kp = 0.5;
ki = 0.1;
kd = 0.5;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
figure
hold on
plot(err1)
hold off
grid on
title('Uchyb regulacji 2')
xlabel('czas[s]')
ylabel('amplituda')
legend('kp = 0.5, kd = 0.5, ki = 0.1')

```



```
clear
licznik1 = 5;
mianownik1 = [1 5 2 1];
licznik2 = 1;
mianownik2 = [2 1 1 0];
licznik = licznik1;
mianownik = mianownik1;
r = 1;
z = 0;
kp = 0.3;
ki = 0.05;
kd = 0.6;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
y2 = simOut.get("y2");
figure
hold on
plot(err1)
plot(y2)
hold off
grid on
title('Uchyb regulacji 1, wielkość regulowana, kp=0.3, ki=0.05, kd=0.6')
xlabel('czas[s]')
ylabel('amplituda')
legend('uchyb', 'wielkosc regulowana')
```



```
data = err1.data;
time = err1.time;
len = length(time);
IAE = sum(abs(time.*data))/len
```

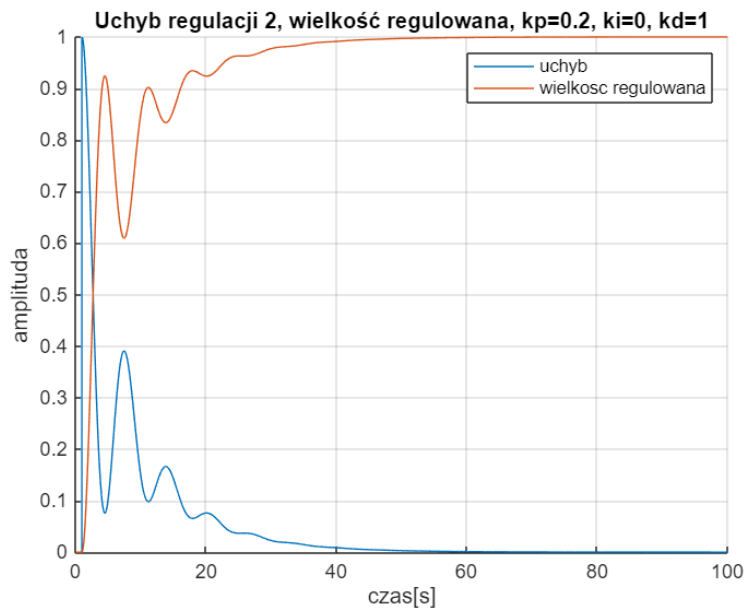
IAE = 0.3648

```
ISE = sum((time.*data).^2)/len
```

ISE = 0.4267

```
clear
licznik1 = 5;
mianownik1 = [1 5 2 1];
licznik2 = 1;
mianownik2 = [2 1 1 0];
licznik = licznik2;
mianownik = mianownik2;
r = 1;
z = 0;
kp = 0.2;
ki = 0;
kd = 1;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
y2 = simOut.get("y2");
figure
hold on
plot(err1)
plot(y2)
hold off
grid on
title('Uchyb regulacji 2, wielkość regulowana,  $k_p=0.2$ ,  $k_i=0$ ,  $k_d=1$ ')
xlabel('czas[s]')
ylabel('amplituda')
```

```
legend('uchyb', 'wielkosc regulowana')
```



Czas regulacji: 23s.

```
data = err1.data;  
time = err1.time;  
len = length(time);  
IAE = sum(abs(time.*data))/len
```

IAE = 0.5056

```
ISE = sum((time.*data).^2)/len
```

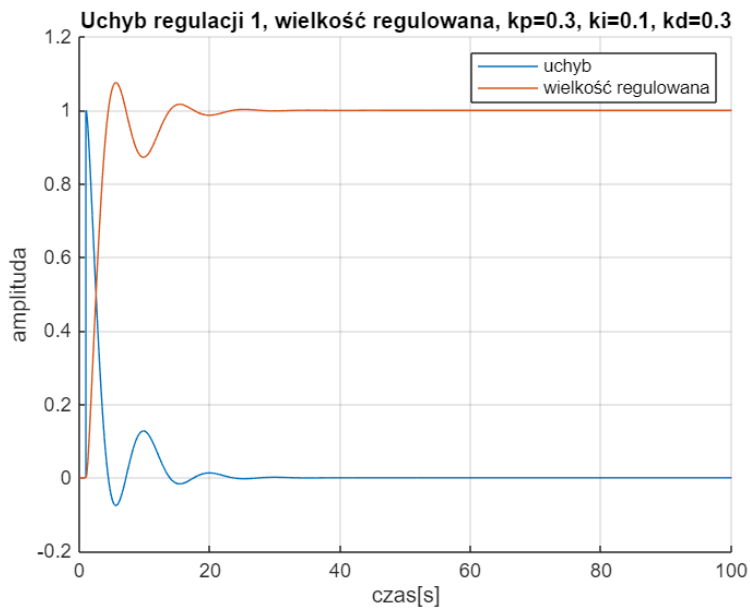
ISE = 0.7249

```
clear  
licznik1 = 5;  
mianownik1 = [1 5 2 1];  
licznik2 = 1;  
mianownik2 = [2 1 1 0];  
licznik = licznik1;  
mianownik = mianownik1;  
r = 1;  
z = 0;  
kp = 0.3;  
ki = 0.1;  
kd = 0.3;  
simOut = sim('regulacja.slx');  
err1 = simOut.get("err");  
y2 = simOut.get("y2");  
figure  
hold on  
plot(err1)  
plot(y2)  
hold off
```

```

grid on
title('Uchyb regulacji 1, wielkość regulowana, kp=0.3, ki=0.1, kd=0.3')
xlabel('czas[s]')
ylabel('amplituda')
legend('uchyb', 'wielkość regulowana')

```



Czas regulacji: 13s.

```

data = err1.data;
time = err1.time;
len = length(time);
IAE = sum(abs(time.*data))/len

```

IAE = 0.1147

```

ISE = sum((time.*data).^2)/len

```

ISE = 0.0978

```

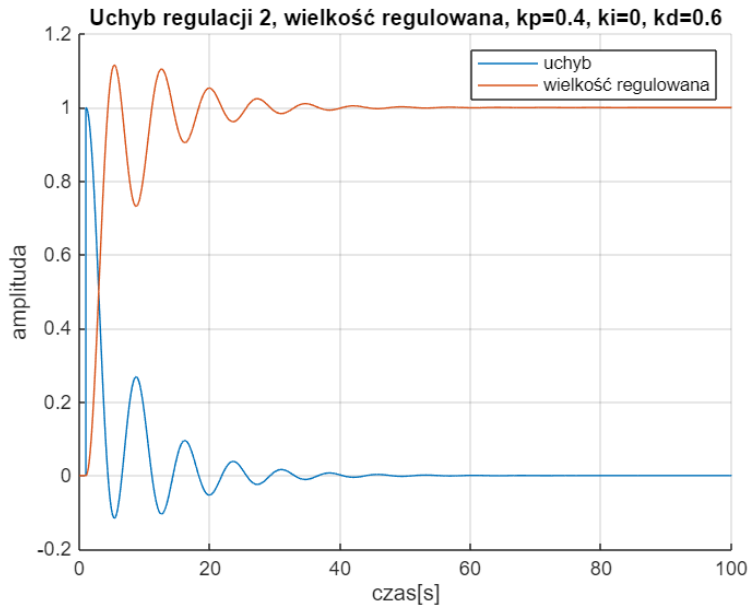
clear
licznik1 = 5;
mianownik1 = [1 5 2 1];
licznik2 = 1;
mianownik2 = [2 1 1 0];
licznik = licznik2;
mianownik = mianownik2;
r = 1;
z = 0;
kp = 0.4;
ki = 0;
kd = 0.6;
simOut = sim('regulacja.slx');
err1 = simOut.get("err");
y2 = simOut.get("y2");
figure

```

```

hold on
plot(err1)
plot(y2)
hold off
grid on
title('Uchyb regulacji 2, wielkość regulowana, kp=0.4, ki=0, kd=0.6')
xlabel('czas[s]')
ylabel('amplituda')
legend('uchyb', 'wielkość regulowana')

```



Czas regulacji: 21s.

```

data = err1.data;
time = err1.time;
len = length(time);
IAE = sum(abs(time.*data))/len

```

IAE = 0.2899

```

ISE = sum((time.*data).^2)/len

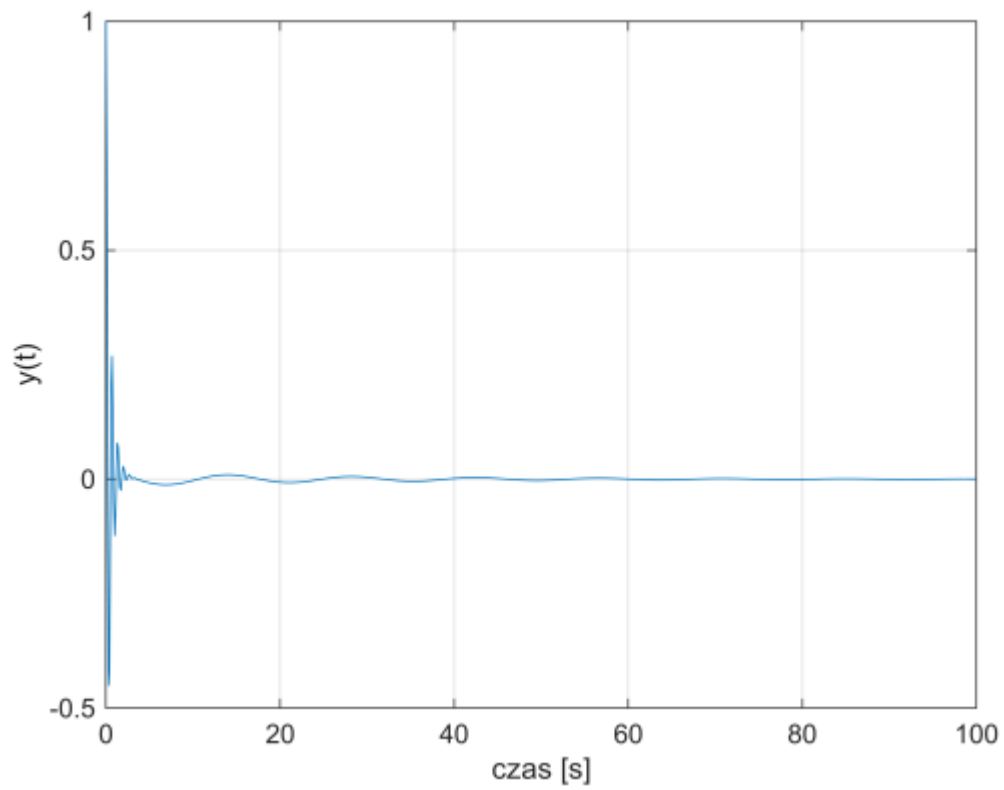
```

ISE = 0.3179

```

r = 1;
z = 0;
kp = 0.2;
ki = 0;
kd = 1;
parametry = [kp, ki, kd];
rst = fminsearch(@funkcje_kosztu, parametry)

```



```
kp = 0.3;  
ki = 0.1;  
kd = 0.3;  
parametry = [kp, ki, kd];  
rst = fminsearch(@funkcje_kosztu, parametry)
```

